Do Now · retrieval

Heuristics, modelling and pipelining

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does a heuristic provide?

Q2

What is throughput?

Q3

What is a pipeline stall?

Q4

Binary search for 13 in indexes0–14 of [0,1,…,14]. List midpoint indexes.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Define the heuristic — A heuristic quickly estimates how promising a choice is; it is not automatically an exact solution.

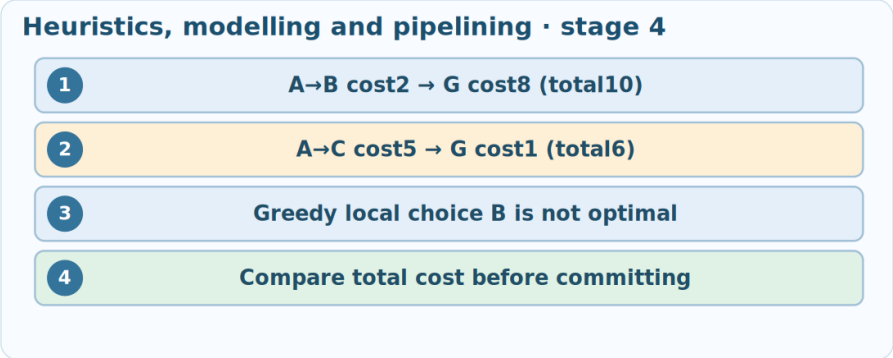
Core idea 2

Model the trade-off — Separate local edge cost, accumulated cost and estimated remaining cost before choosing a path.

Core idea 3

Pipeline and performance models — A pipeline can overlap independent stages, increasing throughput, but dependencies and resource constraints may cause stalls.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

heuristic — Estimate/rule guiding search.

performance model — Simplification estimating resource use.

pipeline — Overlap processing stages.

stall — Delay in progress through stages.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

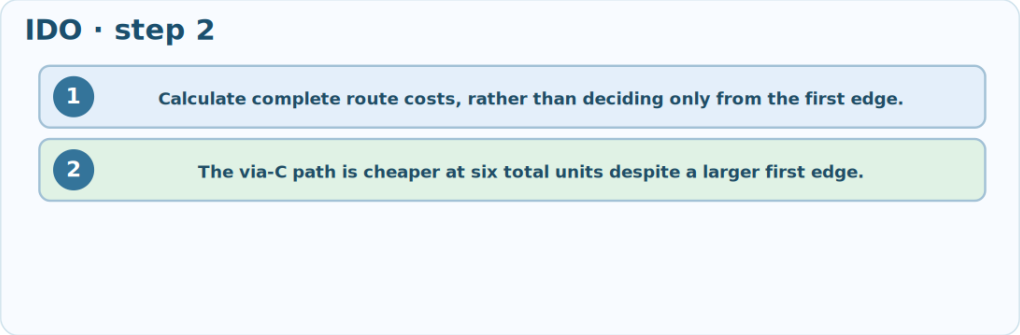
Problem

Evaluate two routes A–B–G (3+7) and A–C–G (5+1).

Follow the worked method

1. Predict first — Calculate complete route costs, rather than deciding only from the first edge.
2. Explain the method — The via-C path is cheaper at six total units despite a larger first edge.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

Task one occupies S1 in cycle 1, with one new task entering each later cycle.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain why choosing the cheapest immediate edge is not guaranteed to produce a shortest path. [4]

Question 2

Five independent jobs pass through 3 one-cycle stages. Give ideal completion cycle for all five and compare with non-overlapped 15 cycles. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate using an A* heuristic and a computational performance model to plan emergency routes. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Choosing the cheapest immediate edge always yields the cheapest complete path.

Use key vocabulary from memory

1. Explain heuristic in your own words.

2. Explain performance model in your own words.

3. Explain pipeline in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.